8.4.2 Equipments

Benedict's solution (fresh; certainly not more than 3 months old), Dropper, Test-tube, Test-tube holder. Spirit lamp, Match box, Kidney tray, containers.

Quality checking of the Benedict's solution

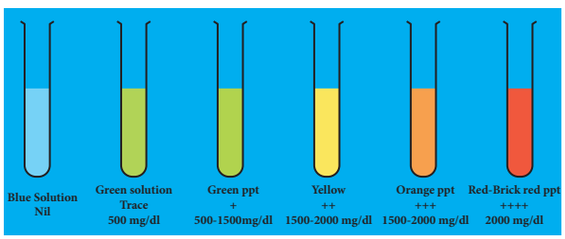
Benedict's solution is blue in color. In order to check purity of Benedict's solution take 5 ml of Benedict's solution in test tube and heat it. If it does not change color, it means, it is pure. Spirit lamp, Match box, Kidney tray, Container.

8.4.3 Procedure

- Take 5 ml (one teaspoon) of Benedict's solution in the test-tube.
- Holding the test-tube with the holder, heat it over a spirit lamp till the Benedict's Solution boils without overflowing.
- Drop 8 to 10 drops of urine into the boiling Benedict's solution.
- After again boiling the mixture, let it cool down.
- While cooling, the mixture changes color.
- Observe the color change and precipitate formation and analyze the test result

8.4.4 Result interpretation

The colour of the mixture serves as a guide to the amount of sugar in the urine :



(ppt – precipitate)

8.4.5 Test For Albumin

- Fill three-fourth of a test tube with filtered urine (filtering removes pus if present).
- See the reaction of the urine is acidic. If found alkaline, add one drop of acetic acid and make it acidic.
- Heat the upper third of the urine over sprit lamp and allow it to boil.
- A cloud may appear either due to phosphate or albumin.
- Add acidic acid drop by drop in to the test tube.
- If the urine is still cloudy it indicates the presence of albumin.
- If it becomes clear it indicates the presence of phosphates.
- No albumin is presence in the normal urine.
- If the urine is highly acidic or highly alkaline, the reading will be false.

8.4.6 Test for Acetone

- Take 5 ml of urine in a test tube and saturate it with ammonium sulphate.
- Add a small crystal of sodium nitroprusside and mix well.
- Slowly run along the side of the test tube liquor ammonia to form a layer.
- Immediate formation of a purple permanganate colored ring at the junction of the two fluids indicates a positive test

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